

# Shrink-fit Heater **HEAT ROBO**

## ■ Specifications

| Shrink-fit Heater | DENJI<br>電磁5000S | DENJI<br>電磁1200S | Baby3000S    | Baby1200S   | Baby1000 |
|-------------------|------------------|------------------|--------------|-------------|----------|
| Spec.             |                  |                  |              |             |          |
| CODE              | HRD-02S          | HRD-01S          | HRB-03ST     | HRB-02S     | HRB-01   |
| Heating method    | Induction type   |                  | Hot air type |             |          |
| Output power      | 5000 w           | 1200 w           | 3000 w       | 1200 w      | 1000 w   |
| Cutter dia.       | ~φ32 [1"]        | ~φ12 [1/2"]      | ~φ32 [1"]    | ~φ12 [1/2"] |          |
| Heating time      | 18 sec.          |                  | 70 sec.      | 120 sec.    | 180 sec. |

## ■ Compatibility table by holder type

| Shrink-fit Heater | DENJI<br>電磁5000S | DENJI<br>電磁1200S | Baby3000S | Baby1200S | Baby1000 |
|-------------------|------------------|------------------|-----------|-----------|----------|
| Holder type       |                  |                  |           |           |          |
| MONO series       | ○                | △                | ○         | △         | ×        |
| 2 PIECE type      | ○                | ○                | ○         | ○         | ○        |
| UNO               | ○                | ○                | ○         | ○         | ×        |
| HYPER VERSION     | ○                | ×                | ○         | ×         | ×        |
| Z                 | ○                | ×                | ○         | ×         | ×        |
| STRAIGHT arbor    | ○                | ▲                | ○         | △         | ×        |

[○]: Available [X]: Not available [△]: Refer to size chart. [▲]: Requires recombinations of adaptors and bases. \*Please verify compatibility for specific holders.

## HOT AIR HEATER

No over-heating.  
Compact Affordable.

### Baby3000S

| CODE        | Area          | Size<br>(W×D×H)   | Power<br>(Single phase) | Kg<br>lbs   |
|-------------|---------------|-------------------|-------------------------|-------------|
| HRB-03ST    | Japan         | 430 × 330 × 660mm | 200V                    | 9.5<br>(21) |
| -03ST-230NA | North America |                   | 230V                    |             |
| -03ST-230EU | Europe        |                   |                         |             |
| -03ST-230AS | Asia          |                   |                         |             |

#### ■ Std. Access.

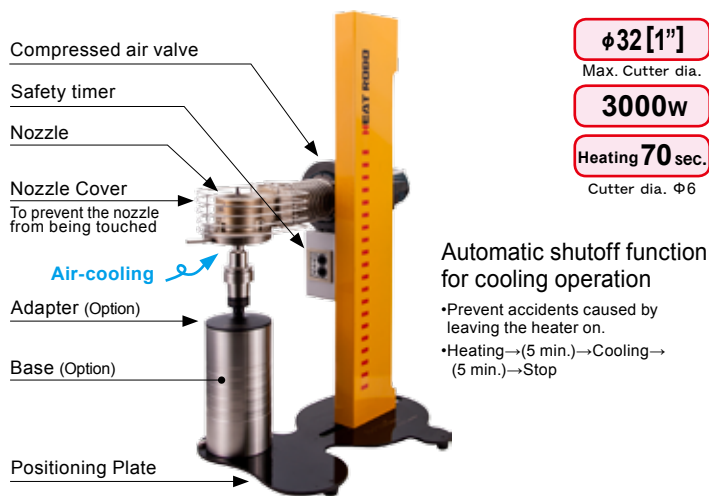
- Safety timer
- Heat-resistant gloves
- Tweezers

#### ■ Option

- Please choose parts on P.12-15

#### ■ Note

- Factory compressed air (5kgf/cm<sup>2</sup>) is required.(consumption air volume: 245 t/min)
- Prepare an air tube (O.D:8mm) and a connection coupling.
- A type without safety timer is also available. **Ex.** HRB-03S



### Baby1200S



φ12 [1/2"]  
Max. Cutter dia.  
1200w  
Heating 120 sec.  
Cutter dia. φ6

| CODE       | Area          | Size<br>(W×D×H)   | Power<br>(Single phase) | Kg<br>lbs   |
|------------|---------------|-------------------|-------------------------|-------------|
| HRB-02S    | Japan         | 370 × 260 × 590mm | 100V                    | 8.0<br>(18) |
| -02S-120NA | North America |                   | 120V                    |             |

#### ■ Std. Access.

- Timer
- Heat-resistant gloves
- Tweezers

#### ■ Option

- Please choose parts on P.12-15

#### ■ Note

- Some holders cannot be used.

### Baby1000



φ12 [1/2"]  
Max. Cutter dia.  
1000w  
Heating 180 sec.  
Cutter dia. φ6

| CODE   | Area  | Size<br>(W×D×H)   | Power<br>(Single phase) | Kg<br>lbs  |
|--------|-------|-------------------|-------------------------|------------|
| HRB-01 | Japan | 340 × 160 × 410mm | 100V                    | 3.5<br>(8) |

#### ■ Std. Access.

- Adapter (ADH-SLK)
- Timer
- Nozzles, 2 pieces (HNZ-17, HNZ-22)
- Heat-resistant gloves
- Tweezers

#### ■ Option

- Please choose parts on P.12-15

#### ■ Note

- A transformer is required in other voltage.  
Please consult MST about the details of the transformer.

# INDUCTION HEATER

Compact desktop type.  
Easy operation with touch panel.

## DENJI 電磁5000S

| CODE     | Size<br>(W×D×H) | Power<br>(3 phases) | Kg<br>lbs  |
|----------|-----------------|---------------------|------------|
| HRD-02SH | 340×470×750mm   | 200-240V            | 30<br>(67) |

### ■Std. Access.

- Heat-resistant gloves •Tweezers

### ■Option

- Heating coil(HRD2-CL) • Please choose parts on P.12-15

### ■Note

- Use type HSB/HSC for cutter stopper.
- Compressed air required: 5Kg/cm<sup>2</sup> (air-consumption volume: 245t/min)
- The customer should prepare the following items : auxiliary parts including the air tube (O.D: 8mm.) an air filter and a coupler.

Heating Coil (Option)

Air-cooling

Adapter (Option)

Base (Option)

Touch Panel

- Timer
- Coil selection
- Heating
- Cooling

Positioning Plate

A3 size

φ32 [1"]

Max. Cutter dia.

5000w

Heating 18 sec.

Cutter dia. Φ6

Heating coil (Option)

| CODE     | Holder I.D                                                        | Heating time | φC     | H  |
|----------|-------------------------------------------------------------------|--------------|--------|----|
| HRD2-CL1 | φ3~ 6 [1/8" ~ 1/4]                                                | 18 sec.      | 25     | 33 |
| -CL2     | φ7~12*1 [5/16" ~ 1/2]                                             | 28 sec.      | 35     | 42 |
| -CL3     | φ16, 20*2*3 [5/8", 3/4]                                           |              | 45     | 47 |
| -CL4     | φ25 [1"]                                                          | 40 sec.      | 55     | 57 |
| -CL5     | MONO series (•MONO 3°[SLRB-SLFB8/10/12/16-M22])                   | 35 sec.      | fig. 1 |    |
| -CL6     | HYPER VERSION (•Short type(All sizes) •Heavy-duty type [φ12, 16]) | 40 sec.      | 60     | 47 |
| -CL7     | φ32, HYPER VERSION (•Heavy-duty type [φ20, 25])                   | 60 sec.      | 70     | 67 |

### ■Std. Access.

- Protection sheet for heating coil →P.15

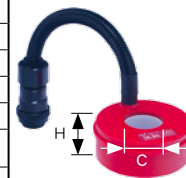
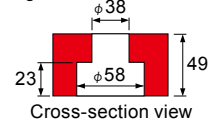
### ■Note

- The coil protection sheet is a consumable part. Replace the coil protection sheet(option) when you see that it is colored or distorted by the heat.

### ■Caution

- To protect the heating coil, don't heat a holder without using a coil protection sheet.
- When ordering, please note that holders for coil CL5, CL6, or CL7 cannot be heated by CL2, CL3, or CL4.
- \*1 Use the coil, No.5 for SLRB-SLFB8/10/12-M22.
- \*2 Use the coil, No.5 for SLRB-SLFB16-M22.
- \*3 Use the coil, No.4 for A50M-SLRB20, F63-SLFB20

fig. 1



## DENJI 電磁1200S

| CODE          | Area          | Size<br>(W×D×H) | Power<br>(Single phase) | Kg<br>lbs  |
|---------------|---------------|-----------------|-------------------------|------------|
| HRD-01S       | Japan         | 230×410×550mm   | 100V                    | 14<br>(31) |
| HRD-01S-120NA | North America | 230×410×550mm   | 120V                    | 19<br>(42) |
| -230AS        | Asia          |                 | 230V                    |            |

### ■Std. Access.

- Heating coil(HRD-CL1, CL2) •Heat-resistant gloves •Tweezers

### ■Option

- Please choose parts on P.12-15

### ■Note

- Use type HSB/HSC for cutter stopper.
- Compressed air required: 5Kg/cm<sup>2</sup> (air-consumption volume: 245t/min)
- The customer should prepare the following items : auxiliary parts including the air tube (O.D: 8mm.) an air filter and a coupler.
- A transformer is required in Europe.
- Please consult MST about the details of the transformer.

Heating Coil (Std.access)

Air-cooling

Adapter (Option)

Base (Option)

Touch Panel

- Timer
- Coil selection
- Heating
- Cooling

Positioning Plate

A4 size

φ12 [1/2"]

Max. Cutter dia.

1200w

Heating 18 sec.

Cutter dia. Φ6

Heating coil (Standard Accessories)

| CODE    | Holder I.D           | Heating time | φC |
|---------|----------------------|--------------|----|
| HRD-CL1 | φ3~ 6 [φ1/8" ~ 1/4]  | 18 sec.      | 18 |
| -CL2    | φ7~12 [φ5/16" ~ 1/2] | 33 sec.      | 28 |

### ■Std. Access.

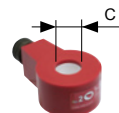
- Protection sheet for heating coil →P.15

### ■Caution

- To protect the heating coil, don't heat a holder without using a coil protection sheet.

### ■Note

- The coil protection sheet is a consumable part. Replace the coil protection sheet(option) when you see that it is colored or distorted by the heat.

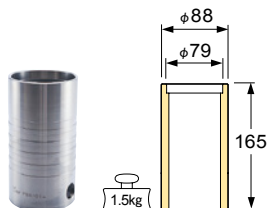


## Option

### Base

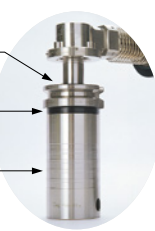
CODE

BAA-01

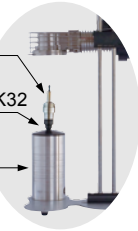


### Examples of typical uses

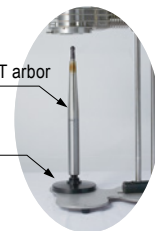
Holder  
BT50  
Adapter  
ADH-50  
Base  
BAA-01



Holder  
HSK-E32  
Adapter  
ADH-HSK32  
Base  
BAA-01



Long  
STRAIGHT arbor  
Adapter  
BAS-01



### Adapter

| CODE          | Dimensions |                                                       | Holder                                   |            | Shrink-Fit Heater model for Use                                               |                                                                               |         |                               |                                                                            |                                                                                               |
|---------------|------------|-------------------------------------------------------|------------------------------------------|------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------|-------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
|               |            |                                                       | Type                                     | Screw size | HRD-02S                                                                       | HRD-01S                                                                       | HRB-03S | HRB-02S                       | HRB-01                                                                     |                                                                                               |
| ADH-SLK       |            | 2 PIECE type<br>SLIMLINE collet<br><br>STRAIGHT arbor | 6 type                                   | M6         | ○                                                                             | ○                                                                             | ○       | ○                             | ○                                                                          | ※Adaptor (ADH-SLK) is included. No need to purchase base (BAA-01) for 2-piece collet type 12. |
|               |            |                                                       | 8 type                                   | M8         |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
|               |            |                                                       | 12 type                                  | M10        |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
|               |            |                                                       | ST10                                     | M6         |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
|               |            |                                                       | ST12                                     | M8         |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
|               |            | ST16/20/25                                            | M10                                      |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| ADH-HSK25     |            | MONO 3°                                               | E25                                      | ○          | △<br>※<br>Some holders cannot be used. Please refer to the dimensional sheet. | ○                                                                             | △       | ※Some holders cannot be used. | ×                                                                          |                                                                                               |
| MONO CURVE    |            | E32                                                   |                                          |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| UNO           |            | E40                                                   |                                          |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| HYPER VERSION |            | A40                                                   |                                          |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| Z             |            | A50 / E50 / F63                                       |                                          |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| -40           |            |                                                       | BT40 / A63<br>F63 / F80PD<br>DN40 / CT40 |            |                                                                               |                                                                               |         |                               | Available dimension<br>C=Max. φ32<br>※M=22 has a max. diameter of 24 at C. |                                                                                               |
| -50           |            |                                                       | BT50 / A100<br>DN50 / CT50               |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| -15TR         |            |                                                       | 15TR3                                    |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| -S20TR        |            |                                                       | RS20 / S20TR                             |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| -BT30         |            |                                                       | BT30                                     |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| BAS-01        |            | STRAIGHT arbor                                        | ST16/20/25                               | M10        | ○                                                                             | △<br>※<br>Some holders cannot be used. Please refer to the dimensional sheet. | ○       | △                             | ※Some holders cannot be used.                                              | ×                                                                                             |
|               |            |                                                       | ST32                                     | M16        |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
|               |            |                                                       | ST42                                     | M24        |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |
| -02           |            |                                                       | Carbide Shank<br>ST○○C                   |            |                                                                               |                                                                               |         |                               |                                                                            |                                                                                               |

[○] : Available

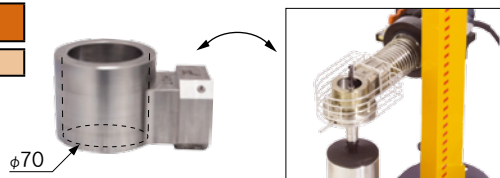
[×] : Not available

[△] : Restricted use

### φ70 Nozzle (Baby3000S)

CODE

HRB-NZL70



### Holder

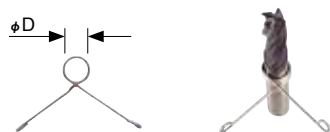
- MONO 3° SLRB32 φ32
- HYPER VERSION Heavy type (All sizes)

## Option

### ■Cutter Stopper

Used as a stopper in the holder when shrink fitting or removing a cutting tool.

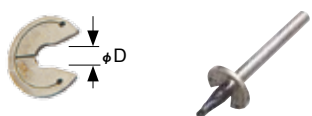
#### HSA type (Coil Spring type)



| CODE                         | φ D                                                                                  | Set Contents (Q'ty)       |
|------------------------------|--------------------------------------------------------------------------------------|---------------------------|
| <b>HSA-D</b><br>[Ex.: HSA-6] | 3, 3.175[1/8], 4, 5, 6, 7, 8, 9, 10, 11, 12, 16, 20, 25, [3/16, 1/4, 5/16, 3/8, 1/2] | Each size 10 ea. /SET     |
| <b>HSA-E</b>                 | 3, 4, 6, 8, 10, 12                                                                   | 1 ea., 6 pieces in total  |
| <b>-F</b>                    | 3, 4, 5, 6, 7, 8, 9, 10, 11, 12                                                      | 1 ea., 10 pieces in total |
| <b>-EF</b>                   | 3, 4, 5, 6, 8, 10, 12, 16, 20, 25                                                    | 1 ea., 10 pieces in total |

■Note • It cannot be used with HEAT ROBO DENJI.

#### HSB type (Plate Spring type)

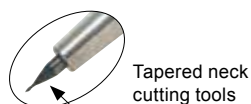


| CODE                         | φ D                                                                                                          | Set Contents (Q'ty)       |
|------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>HSB-D</b><br>[Ex.: HSB-3] | 3, 3.175[1/8], 4, 5, 6, 7, 8, 9, 10, 11, 12, 16, 20, 25, 32 [3/16", 1/4", 5/16", 3/8", 1/2", 5/8", 3/4", 1"] | 1 piece                   |
| <b>-E</b>                    | 3, 4, 6, 8, 10, 12                                                                                           | 1 ea., 6 pieces in total  |
| <b>-F</b>                    | 3, 4, 5, 6, 7, 8, 9, 10, 11, 12                                                                              | 1 ea., 10 pieces in total |
| <b>-EF</b>                   | 3, 4, 5, 6, 8, 10, 12, 16, 20, 25                                                                            | 1 ea., 10 pieces in total |

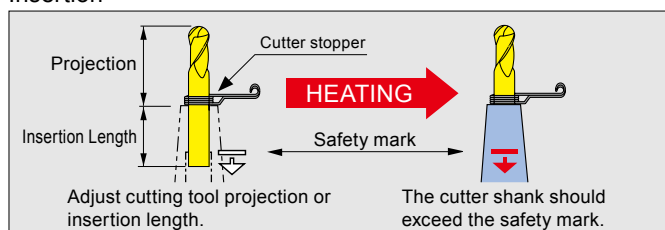
#### HSC type (Slit Collet type)



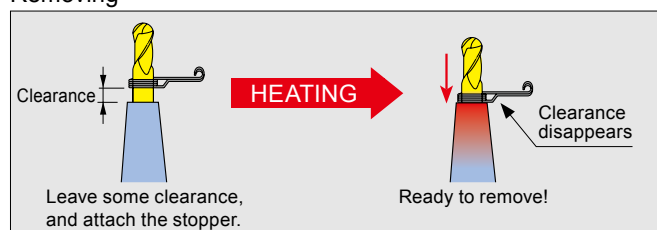
| CODE                         | φ D                 |
|------------------------------|---------------------|
| <b>HSC-D</b><br>[Ex.: HSC-3] | 3, 3.175[1/8], 4, 6 |



#### Insertion



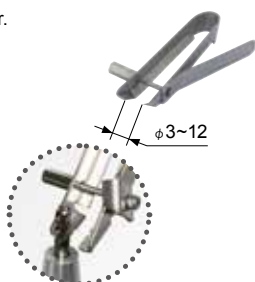
#### Removing



#### Cutter Pliers

Easy to hold a cutter.

| CODE          |
|---------------|
| <b>HPY-01</b> |



#### Stopper Pliers

Easy to attach and detach for HSB type.

| CODE          |
|---------------|
| <b>SPY-01</b> |



#### Heat-Resistant Gloves

For additional order

| CODE          | NOTE       |
|---------------|------------|
| <b>HTB-01</b> | —          |
| <b>-01-R</b>  | Right Hand |
| <b>-01-L</b>  | Left Hand  |



#### Cutter Tray

Cooling Tray for heated cutting tools after removal from a holder.

| CODE          |
|---------------|
| <b>SDH-01</b> |

Size:170×170(mm)

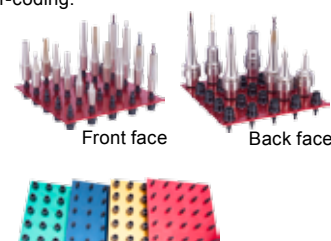


#### Holder Stand

The stand for the SLIMLINE collet, straight arbor and small shank holders (HSK-E25 and E32) is available in four colors, enabling simple color-coding.

| CODE           | Color | Front face                        | Back face                                 | max. Q'ty         |
|----------------|-------|-----------------------------------|-------------------------------------------|-------------------|
| <b>SDKT-RE</b> | Red   | SLIMLINE collet<br>Straight arbor | Small shank<br>holders<br>(HSK-E25 / E32) | 25 pieces<br>each |
| <b>-BL</b>     | Blue  |                                   |                                           |                   |
| <b>-GR</b>     | Green |                                   |                                           |                   |
| <b>-GD</b>     | Gold  |                                   |                                           |                   |

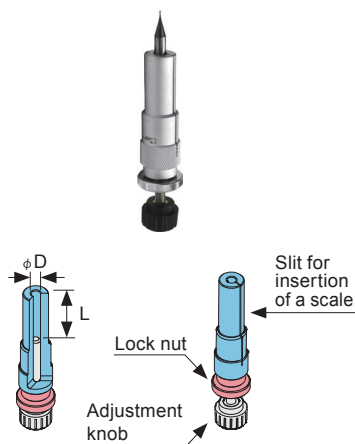
Size:190×190(mm)



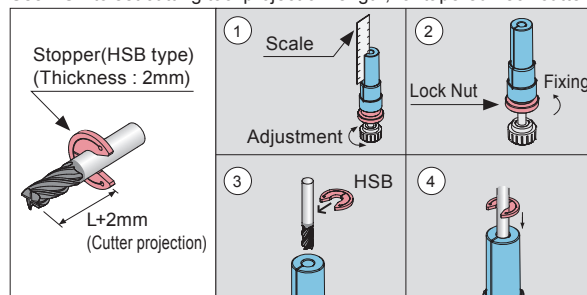
## Cutter Adjuster

Simply adjustment for cutting tool projection and insertion length.

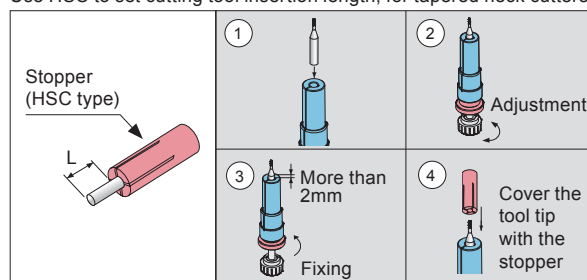
| CODE   | φ D         | L       |
|--------|-------------|---------|
| HAJ-3  | 3           | 10~ 30  |
| -3.175 | 3.175 [1/8] |         |
| -4     | 4           | 13~ 30  |
| -3/16  | [3/16]      |         |
| -6     | 6           | 19~ 45  |
| -1/4   | [1/4]       |         |
| -8     | 8 [5/16]    | 21~ 55  |
| -10    | 10 [3/8]    | 22~ 70  |
| -12    | 12          | 23~ 85  |
| -1/2   | [1/2]       |         |
| -16    | 16 [5/8]    | 26~ 90  |
| -20    | 20 [3/4]    | 37~ 100 |
| -25    | 25          | 40~ 100 |
| -1"    | [1"]        |         |



Use HSB to set cutting tool projection length, for tapered neck cutters.



Use HSC to set cutting tool insertion length, for tapered neck cutters.



## ■Cleaning tools

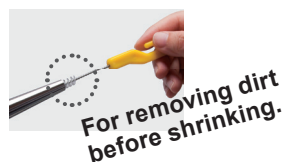
### Brush type

Nylon brushes for internal bores cleaning.

| CODE       | I.D                                                                               |
|------------|-----------------------------------------------------------------------------------|
| AQC-BR-SET | 3 [1/8], 4 [3/16], 6 [1/4], 8 [5/16], 10 [3/8], 12 [1/2], 1 ea. 6 pieces in total |
| BR 3-5     | 3 [1/8 ], 5 pcs.                                                                  |
| BR 4-5     | 4 [3/16], 5 pcs.                                                                  |
| BR 6-5     | 6 [1/4 ], 5 pcs.                                                                  |
| BR 8-5     | 8 [5/16], 5 pcs.                                                                  |
| BR10-5     | 10 [3/8 ], 5 pcs.                                                                 |
| BR12-5     | 12 [1/2 ], 5 pcs.                                                                 |

### ■Caution

- Do not use when holder is hot.



### Rubber grinding stone type

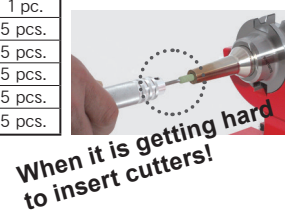
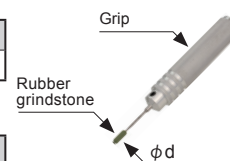
To eliminate strong oxidized film, burned-in dust and oil that have been generated in the internal bores.

### Standard set

| CODE       | Note                 |
|------------|----------------------|
| CLT-GTA-01 | 1 ea. of below model |

### Parts

| CODE        | NAME               | φd | Q'ty   |
|-------------|--------------------|----|--------|
| CLT-GTA-GP  | Grip               | -  | 1 pc.  |
| CLT-GTA 3-5 | Rubber grind stone | 3  | 5 pcs. |
| -GTA 4-5    |                    | 4  | 5 pcs. |
| -GTA 6-5    |                    | 6  | 5 pcs. |
| -GTA 8-5    |                    | 8  | 5 pcs. |
| -GTA10-5    |                    | 10 | 5 pcs. |



## ■Aqua Cool

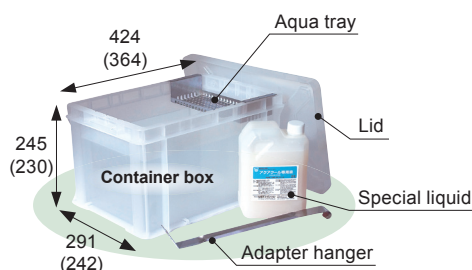
### Aqua cool kit

Cooling kit for cooling SLIMLINE holders after shrink fitting or removing

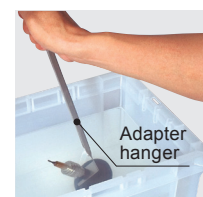
| CODE       | Note                  |
|------------|-----------------------|
| AQC-KIT-01 | 1 ea. for below model |

### Parts

| CODE        | Name                  | Note                                                     |
|-------------|-----------------------|----------------------------------------------------------|
| AQC-EK-01-2 | Special Liquid        | 2ℓ of undiluted Solution, Use at a dilution ratio of 3%. |
| -AT-01      | Aqua Tray             | Drainer plate                                            |
| -AH-01      | Adapter Hanger        | Hanger for lowering holders into liquid.                 |
| CN -245     | Container Box         | -                                                        |
| -FT         | Lid for Container Box | -                                                        |



( ) Inside dimension



Protection sheet for heating coil (For Induction Heater)

Don't heat holders without using a coil protection sheet.

| CODE                                                                                                  | Heater model             |
|-------------------------------------------------------------------------------------------------------|--------------------------|
| HRD -DN <br>Coil No. | HEAT ROBO<br>DENJI 1200S |
| HRD2-DN <br>Coil No. | HEAT ROBO<br>DENJI 5000S |

